FIRE SAFETY INSPECTION REPORT

Lakhsma Innerwear Ltd

ACCORD ID: 23359

South vangnahati, Sreepur, Gazipur

Other Factories: None



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Inspected By: Md.Mahmudur Rahman & Monsurul Akram Azim
Date: 15 November, 2018

Fire Safety Inspection Report

Factory Name: Lakhsma Innerwear Ltd.

Introduction:

Lakhsma Innerwear Ltd. was surveyed for fire safety on 15 November, 2018 by the Accord Inspection Team. The purpose of the survey was to identify significant fire safety issues and to provide recommendations for remediation based on applicable standards specified by the Accord. The scope of this initial fire safety inspection was limited to the review and identification of major fire safety issues. The inspection did not include identification of minor deficiencies, which will be further addressed as part of follow-up inspections.

Limitations

The information in this fire safety inspection report was obtained during a visit to the facility and during interviews with local factory management. It has not been possible to provide independent verification for all the information and data collected, and, therefore, The Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.
The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

General Factory Information

Factory Building name(s):	1. Factory Building- 3 storied(Proposed 4 storied) 2. Shed Bld (Sewing) 3. Utility Bld- Single story 4. Shed Bld 2 (Child care & Medical)

Factory Address:

South vangnahati, Sreepur, Gazipur

Points of contact:

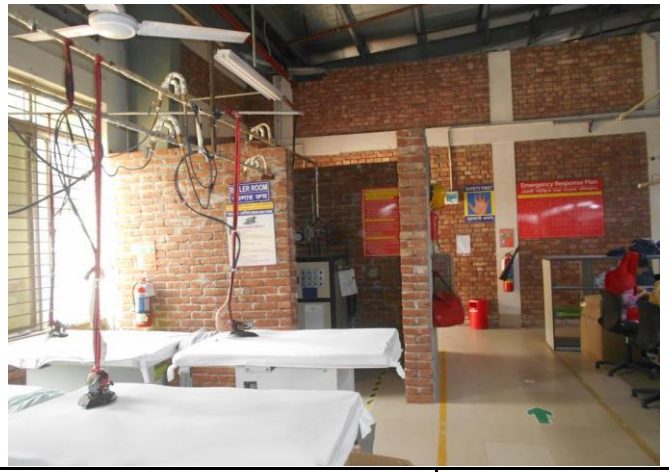
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Findings & Recommendations :

The following tables summarize the major fire safety issues identified during the inspection. Recommendations have been provided to address each issue.
An implementation schedule shall be developed by the factory to remediate each of the findings. The specific timing of improvements, including any requested extensions due to design / installation constraints, shall be submitted to the Accord for approval.

Table 1: Factory Building				
Sl No.	***Category***	***Findings***	***Required Action***	***Remediation Time Frame***
F-1	Storage Areas	Areas used for in-process storage of combustible materials are open to the surrounding occupancy	Provide defined storage areas and limit the storage arrangement as follows:- 1. **General Condition:** Maximum height of 2.4m and maximum area of 23m2. Separate areas of unenclosed combustible storage by a minimum clear distance of 3m. 2. **If sprinkler protected:** maximum height of 3.66m and maximum area of 93m2. Separate areas of unenclosed combustible storage by a minimum clear distance of 3m. or 3. Enclose the storage area from the surrounding occupancy with a minimum 1 hour construction.	***Within 3 months***

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F-2	***Fire Rated Construction***	Boiler units are located inside the staircase enclosure without any fire separation.	Provide minimum 1-hr fire rated construction to separate the boiler room from other occupancies.	***Within 3 months***

				
F-3	***Fire Rated Construction***	Dining area are not separated by proper fire rated construction. Location: 2nd floor	Separate the dining rooms by minimum 1-hr fire rated construction from any other occupancy. The opening protective shall also be minimum 1-hr fire rated. Also ensure direct access to exit stair from dining.	***Within 3 months***





F-4	***Egress***	Aisles were found obstructed by process goods/sewing tables/finished cartons at several locations throughout the factory.	Aisles shall always be free of any kind of obstruction. Provide minimum clear aisle widths of 36-inch.	***Immediate***







F-5	***Egress***	Building did not have stair designation signs and occupant load signs at the required locations.	Provide stair designation signs and occupant load signs at a conspicuous space near floor entries.	***Within 1 month***
				
F-6	***Egress***	Egress doors/gates with locking features not complying with NFPA101 are provided at building exits at ground & 1st floor.	Remove locking features from all egress doors/gates. If locks are required for security reasons, utilize specialized door locking features complying with NFPA101.	***Immediate***

				
F-7	***Fire Rated Construction***	Penetrations through fire rated floor/staircase assemblies were found unsealed.	Seal all penetrations and openings in floor/staircase (full/required thickness) assemblies by listed fire rated materials to maintain the required fire resistance rating	***Within 3 months***





F-8	***Fire Rated Construction***	Penetrations / Openings through fire rated floor/ceiling assemblies were found unsealed. Location: Ground / 1st floor	Seal all penetrations and openings in floor/ceiling (full/required thickness) assemblies by listed fire rated materials to maintain the required fire resistance rating	***Within 3 months***







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F-9	***Fire Protection System***	The factory building has an occupied floor (as per architectural drawing) greater than 10 meter (33 feet)	Design & install a new class III standpipe system & fire department (Siamese) connection in accordance with NFPA 14. The standpipe system shall be connected to a dedicated fire pump meeting minimum hydraulic demands of NFPA 14 & Accord standard at the required points. Installation of the fire pump shall be as per NFPA 20. Submit the design to Accord for review and approval before installation. Upon approval & installation the owner shall contact the Accord prior to conducting the final acceptance testing of the standpipe system installation. Also inspect, test and maintain the standpipe system & fire pumps, and keep written records on-site, in accordance with NFPA 14, 20 and ACCORD standards.	***Design within 3 months; Install within 6 months.***
F-10	***Fire Protection System***	The factory has been installed sprinkler system in factory building, But not installed as per standard.	Factory need to maintain the sprinkler system as per standard/ NFPA-13.	***Design within 3 months; Install within 6 months.***



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F-11	***Fire Protection System***	Fire pumps are provided and needed to meet as per Accord Building Standard 5.5.1	Provide fire pumps. Installation of the fire pumps shall be required to provide shop drawing & hydraulic calculations as outlined in NFPA 13, 20, 22 and 24. These drawings shall include all details as outlined in NFPA 13, 20, 22 & 24. All fire protection water supply system installation design documents shall be submitted for review by the Chief Safety Inspector of Accord prior to commencement of installation. Upon approval & installation the owner shall contact the Accord prior to conducting the final acceptance testing of the installation to allow the Accord the option of witnessing this esting & conduct a final inspection of the installation.	***Design within 3 months; Install within 6 months.***

F-12	***Fire Rated Construction***	The Compressor room is not separated by fire rated construction with pump room. (openings/penetrations/non rated ceilings).	Separate the Compressor room by a minimum 2-hr fire rated construction along with a Type-I (BNBC 2006) ceiling construction. Seal and/or protect all openings to maintain fire separations with pump room.	***Within 3 months***



F-13	***Egress Lighting***	Several exits are not provided with exit signage's. Directional exit signs were missing at the change of exit direction.	Provide illuminated exit signs at every floor and room exits. Provide illuminated directional exit signs where the change of direction is not obvious.	***Within 1 month***



Table 2: Shed 1 (Sewing shed)

Sl No.	***Category***	***Findings***	***Required Action***	***Remediation Time Frame***

F-14	Storage under cutting table	Storage was located under the cutting table. Location: Ground floor.	Remove all storage from under the cutting table..	***Immediate***
				
F-15	Fire Rated Construction	Egress doors/gates with locking features/slide bolts not complying with NFPA101 are provided at exits.	Remove locking features/slide bolts from all egress doors/gates. If locks are required for security reasons, utilize specialized door locking features complying with NFPA101.	***Immediate***



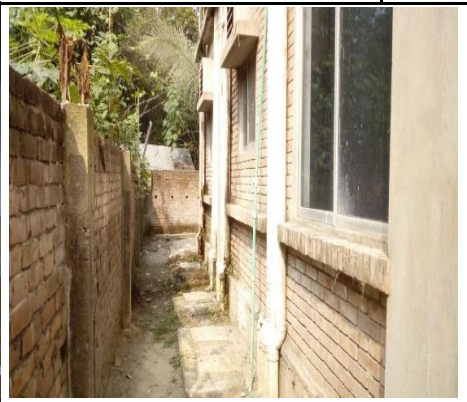
F-16	***Fire Rated Construction***	Room used for combustible storage is open to the dining area and not separated by proper fire rated construction.	Separate the storage room by minimum 1-hr fire rated construction from any other occupancy. The opening protective shall also be minimum 1-hr fire rated.	***Within 3 months***



Table 3: All Occupancies				
Sl No.	**Category**	**Findings**	**Required Action**	**Remediation Time Frame**
F-17	**Fire Detection & Alarm**	The fire detection system has been provided by the factory but drawing is not reviewed by the Accord & Installation shall be verified by the Accord.	Design and install an addressable fire detection & alarm (Audio & Visual) system replacing the existing one in compliance with NFPA 72 & Accord standard. Submit the design to Accord for review and approval before installation. Upon approval & installation the owner shall contact the Accord prior to conducting the final acceptance testing of the fire detection & alarm system installation.	**Design within 4 months; Installation within 9 months**
F-18	**ITM Fire Detection & Alarm**	Inspection, testing, and maintenance for the fire detection & alarm system is not in accordance with NFPA 72.	Inspect, test and maintain the fire detection & alarm system, and keep written records on-site, in accordance with NFPA 72	**Within 3 months**

F-19	***ITM Fire Protection***	Inspection, testing and maintenance of portable fire extinguishers is not in accordance with NFPA 10.	Inspect, test and maintain the portable fire extinguishers and keep written records onsite, in accordance with NFPA 10.	***Within 3 months***
F-20	***Occupancy load***	Building did not have occupant load signs at required locations.	Provide occupant load signs at required locations.	Immediately
F-21	Manual On/Off Switch	Manual on-off switch is provided for emergency lighting units.	Remove manual on/off switches from emergency lighting units to prevent them from being switched off.	***Within 1 months***
				

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F-22	***Egress Court/ Unprotected Exterior Discharge Path***	The exit discharge path is less than 10 ft. wide and is not separated from the building interior.	Seal all penetrations and openings to the interior of the building along the discharge path, up to a height of 10 ft. to provide a minimum 1-hr fire separation.	***Within 3 months***
	  			
F-23	***Egress Lighting***	Based on the number and location of emergency lights observed, adequate illumination levels are not anticipated along egress routes.	Test the emergency lighting system on each floor and provide additional emergency fixtures to provide adequate illumination along the means of egress. Provide a minimum illumination of 10 lux at the floor level within exit stairs and exit discharge paths and minimum 2.5 lux along exit access aisles.	***Within 3 months***